



WhatsApp CRM

A multi-business WhatsApp sales & support platform: one team inbox, lead management, bulk campaigns, chatbot and Facebook-ad lead tracking, all on the official WhatsApp Cloud API.

Team Inbox

Leads & Pipeline

Bulk Campaigns

Chatbot

Facebook Ads

Multi-business SaaS

Product documentation · Version 1.0 · October 2026

Covers: purpose, problems solved, business model, roles & permissions, every feature, technology, architecture, security, setup and roadmap.

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In one sentence: WhatsApp CRM lets you (the platform owner) sell a monthly WhatsApp business tool to many companies. Each company connects its own WhatsApp number, and its team uses it to reply to customers, sort leads, run bulk campaigns and automate replies, with every company's data kept separate.

At a glance

3 login roles

Super Admin (platform owner), Admin (business owner), Agent (business staff).
Customers/leads are contacts, not logins.

1 number per business

Each business connects one official WhatsApp number (Meta Cloud API). Unlimited team members can share it, within the plan limit.

Monthly plans

Plans set limits on agents, contacts and messages per month. The Super Admin renews subscriptions and can suspend a business.

1. Product overview

WhatsApp is where Indian customers reply fastest, but most businesses manage it from one phone. Messages get missed, nobody knows which lead is interested, and sending offers to hundreds of people means forwarding one by one or risking a ban with unofficial tools.

WhatsApp CRM turns one official WhatsApp Business number into a shared, organised system for the whole team:

- **Team Inbox:** every chat in one screen, on any number of computers at once, with chats auto-assigned to agents.
- **Leads & pipeline:** every person who messages becomes a lead with a status (New, Interested, Converted, Not interested...), tags, notes, follow-up reminders and custom details.
- **Bulk campaigns:** send approved WhatsApp templates to a list, a status group, a tag or an Excel sheet, now or at a scheduled time, with delivery and read reports.
- **Chatbot:** answers common questions 24x7 with a menu, collects lead details, and hands the chat to a person when needed.
- **Facebook / Instagram ads:** leads from Click-to-WhatsApp ads are tagged with the ad they came from, so you can see which ad brings leads and conversions.
- **Multi-business SaaS:** the platform owner (Super Admin) hosts many businesses, each with its own number, team, data and monthly plan.

Main modules

Module	What it does
Dashboard	Today's chats, new leads, 7-day message chart, lead pipeline, follow-ups due, leads from ads, plan usage, team performance (admin).
Inbox	Real-time chat with customers: text, media, templates, quoted replies, reactions, internal notes, assignment, resolve, lead status and follow-ups.
Contacts / Leads	Lead list with status tabs, filters, bulk actions, Excel/CSV import & export, follow-ups, custom fields.
Bulk campaigns	Template broadcasts to chosen audiences, scheduling, editing before start, pause/resume/cancel, per-recipient report.
Templates	Create WhatsApp message templates, submit to Meta, track approval, edit approved templates, set variable defaults.
Chatbot	Welcome menu, keyword replies, lead questions, business hours, hand-off to the team.
Team	Add agents, disable them, reset passwords; see each member's open chats.
Settings	WhatsApp connection, automation, opt-out words, lead statuses, contact fields, plan usage, sandbox tester.
Super Admin panel	Businesses, plans, subscriptions, suspend/activate, "login as admin", platform stats and audit logs.

2. Problems it solves

Problem today	How WhatsApp CRM solves it
One phone, many staff: only one person can reply, and chats are lost when they are busy or on leave.	Shared team inbox on any number of computers at once. Chats are auto-assigned round-robin and can be transferred. Sound + desktop alerts for new messages.
No idea which leads are interested, converted or lost.	Lead statuses (fully customisable), status tabs and counts, dashboard pipeline. Status also changes automatically when a customer writes "interested" / "not interested".
Leads are forgotten after the first chat.	Follow-up date + note on each lead; "Follow-ups due" on the dashboard and badges in the inbox.
Sending offers to hundreds of customers means manual forwarding, or a ban risk with unofficial tools.	Bulk campaigns on the official Cloud API using Meta-approved templates, from a list, tag, status, ad or uploaded Excel sheet, with delivery/read reports.
Same questions (fees, timing, address) answered again and again.	Chatbot with a menu and keyword replies, available 24x7; hands over to a person when needed.
Money spent on Facebook/Instagram ads but no idea which ad works.	Click-to-WhatsApp ad leads are tagged with the ad; dashboard shows leads and conversions per ad.
No visibility into staff performance.	Team report: chats handled, messages sent, leads converted, median reply time (7 / 30 days).
Customer data scattered across phones and sheets.	One database per business with custom fields, import from Excel, export to Excel/CSV.
Customers who don't want messages keep getting them.	Opt-out keywords (STOP, UNSUBSCRIBE...) automatically exclude them from campaigns; START opts them back in.
For the platform owner: building a separate tool for each client.	One multi-business platform: add a business in minutes, set its plan, and every business's data stays isolated.

3. Who it is for

Any business that gets enquiries on WhatsApp and has more than one person handling them. Examples:

Coaching & training institutes

Course enquiries from ads, fee/timing questions answered by the chatbot, counsellor follow-ups, batch announcements by bulk campaign.

Real estate

Property enquiries from ads, site-visit follow-ups, new project launches to "Interested" leads.

Clinics & diagnostics

Appointment questions, reminders (utility templates), health-camp campaigns.

Retail & D2C brands

Order questions, offers to tagged customers, festive campaigns.

Travel & events

Package enquiries, itinerary updates, last-minute offers.

Agencies & resellers

Run the platform for many client businesses from one Super Admin panel.

4. Roles & permissions

Super Admin

Platform owner

Runs the platform. Creates businesses, plans and subscriptions, suspends/activates businesses, logs in as a business admin for support, sees all audit logs. Does not chat with customers.

Admin

Business owner

Full control of one business: WhatsApp number, team, templates, campaigns, chatbot, settings, all chats and reports.

Agent

Business staff

Handles customer chats: sees chats assigned to them plus unassigned ones, replies, updates leads. Cannot change business settings.

Customers / leads ("User") are not logins. They are contacts stored inside one business. Every business sees only its own contacts, chats and campaigns.

Permission matrix

Action	Super Admin	Admin	Agent
Platform			
Create / edit / delete businesses	✓	—	—
Create and edit plans, renew subscriptions	✓	—	—
Suspend / activate a business	✓	—	—
Log in as a business admin (support), reset admin password	✓	—	—
Platform stats and audit logs of all businesses	✓	—	—
Chats (Inbox)			
See chats	via login-as	All	Assigned to them + unassigned
Reply, send media and templates, start a new chat	via login-as	✓	✓
Assign / transfer chats, resolve, take over from the bot	via login-as	✓	✓
Internal notes (edit/delete own notes)	via login-as	✓ (any note)	✓ (own)
Delete own sent message (within 48 h)	via login-as	✓	✓
Hide any message from the CRM	via login-as	✓	—
Contacts / Leads			
View, add and edit contacts, change lead status, tags, follow-ups, custom fields	via login-as	✓	✓
Bulk change status / add tags	via login-as	✓	✓
Delete contacts (single or bulk)	via login-as	✓	—
Import Excel/CSV, export Excel/CSV	via login-as	✓	—
Campaigns, templates, chatbot			
Bulk campaigns (create, schedule, edit, pause, cancel)	via login-as	✓	Only if Admin allows

Create / submit / edit / delete templates, set variable defaults	via login-as	✓	View only
Configure chatbot	via login-as	✓	—
Team & settings			
Add / disable agents, reset their passwords	via login-as	✓	—
Connect WhatsApp number, automation, opt-out words, lead statuses, contact fields	via login-as	✓	—
Dashboard	Platform dashboard	Whole business + team report	Own chats & follow-ups
Change own password, sound/desktop alerts	✓	✓	✓

"via login-as": the Super Admin can open a business as its admin for support. Every action taken this way is shown in a banner and recorded in the audit log as "via Super Admin". The session lasts 2 hours.

5. Features in detail

5.1 Dashboard

- Open, pending and unassigned chats; new leads today; total contacts.
- Messages received and sent in the last 7 days (chart).
- Lead pipeline with counts per status, linking to the filtered contact list.
- Follow-ups due today (agents see their own), leads from ads in the last 30 days with conversions.
- Plan usage this month (messages used / limit) and recent campaigns.
- **Team performance** (admin, 7 or 30 days): chats handled, open chats, messages sent, leads converted, median reply time per member, plus chatbot stats.

5.2 Team Inbox

- Real-time chat list with filters: Open / Pending / Resolved / All, assigned to me, unassigned, bot handling, by agent, by lead status, leads from ads, and search.
- Send text, images, documents, video, audio and approved templates. The **24-hour window** is shown; outside it only templates can be sent (a WhatsApp rule).
- Quoted replies (both directions), customer reactions, internal notes visible only to the team.
- Delete own message within 48 hours (removed from the CRM; WhatsApp has no "delete for everyone" in the API), "Send correction" to resend a fixed text, admin can hide any message. All deletes are soft and audited.
- Header tools: **lead status dropdown**, assign to agent, resolve/reopen, **Details** panel.
- Details panel: name, ad source, lead status, follow-up with note, tags, custom fields, notes, assignment, chat status, chatbot control.
- "🔔 From ad" card on messages that came from a Click-to-WhatsApp ad.
- Sound + desktop notification for new customer messages (per browser on/off).

5.3 Contacts / Leads

- Status tabs with counts; filters for search, tag, source (ads), ad, follow-up (due / overdue / upcoming).
- Inline status change; edit modal with name, phone, email, status, follow-up, tags, notes, custom fields, opt-out.
- Select rows or "all N matching" → change status, add tags, delete, or **send a bulk message** to them.
- **Import** Excel (.xlsx) or CSV up to 10 MB / 50,000 rows: automatic column matching, preview, country code added (default 91), every import gets a batch tag, and a "Send bulk WhatsApp message" button for that batch.
- **Export** to Excel or CSV with the current filters (admin).

5.4 Lead statuses & automation

- Default statuses: New, Contacted, Interested, Converted, Not interested. The admin can rename, recolour, reorder, add (e.g. "Call back") or remove them (Settings → Lead statuses).
- **Auto lead status:** if the customer writes "interested", "haan interested hu", "interest hai" → Interested; "not interested", "interest nahi hai", "no thanks", "nahi chahiye" → Not interested. Questions and long messages are ignored, Converted leads are never changed, and a "🤖 Auto update" note is added in the chat. Can be turned off in Settings.
- Who changed the status and when is stored; this feeds the team report.

5.5 Bulk campaigns

- Audience: all contacts, by lead status, by tags, from a specific ad, hand-picked contacts, or the batch from an uploaded sheet. The number of recipients is shown before sending; opted-out contacts are always excluded.
- Template + variables: each `{{1}}`, `{{2}}` is filled from a contact field (name, phone, email or any custom field) or fixed text. Defaults come from the template and can be changed per campaign. A live preview is shown.
- Send now or **schedule** for a date/time; save as draft.
- **Edit** draft and scheduled campaigns (name, template, audience, variables, time) until 1 minute before start; "Unschedule" turns it back into a draft.
- Pause, resume and cancel; report per recipient: sent, delivered, read, failed, skipped. The worker sends at a controlled speed, survives restarts, and pauses automatically when the plan's monthly limit or the subscription runs out.

5.6 Templates

- Create a template (name, category: Marketing / Utility / Authentication, language, header, body, footer) and submit it to Meta. Status updates (pending, approved, rejected with reason) arrive live.
- **Variables section:** for each `{{n}}` choose what fills it and an example for Meta's review. Stored only in the CRM, so it can be changed any time without re-approval.
- **Edit approved templates:** header, body and footer only (name, category and language are locked by WhatsApp). The edit goes back to review, with Meta's limits enforced (1 edit per 24 hours, 10 per 30 days); blocked while a running or scheduled campaign uses the template.

5.7 Chatbot (rule-based)

- Welcome message with a menu: up to 3 options as reply buttons, up to 10 as a list. Customers can tap, type the number, or type part of the option name.
- Each option can reply with a message, ask lead questions, or hand the chat to the team; options can add a tag.
- **Keyword replies** (contains / exact) for free-typed questions like "fees", "address".
- **Lead questions** save answers into the contact (name, email or custom fields) and mark the lead.
- Business hours with an offline message; "Main Menu" button after answers; restart after silence or after a resolved chat; loop protection.
- The bot stops as soon as a team member replies or takes over. Numbered markers on the settings page match the live WhatsApp-style preview.
- Super Admin can switch the chatbot module on or off per plan.

5.8 Facebook / Instagram Click-to-WhatsApp ads

- When a customer starts a chat from an ad, WhatsApp sends the ad details. The contact gets source "ad", the ad headline and ID (first touch) and the tag `facebook-ad`.
- Filter leads by ad, send campaigns to leads of one ad, and see leads and conversions per ad on the dashboard.

5.9 Contact fields

- Settings → Contact fields: the admin adds, renames or deletes custom fields (Course, City, Fees...). Name, phone and email are built in.
- Fields are used in template variables, campaigns and chatbot questions, and filled on each contact (Contacts → Edit, Inbox → Details).
- Extra sheet columns and new chatbot fields are added automatically. A field used by a template, an active campaign or the chatbot cannot be deleted (the screen shows where it is used).

5.10 Team & multi-computer use

- Admin adds agents (within the plan limit), disables them (their chats return to the unassigned queue) and resets passwords.
- Auto-assignment gives new chats to active agents one by one (round-robin).
- Everyone can be logged in on many computers at once; all screens update live.

5.11 Settings

- WhatsApp number: connect Phone number ID, WABA ID and access token (encrypted), test, disconnect. Sandbox mode until connected.
- Automation & permissions: auto-assign, allow agents to run campaigns, auto lead status, opt-out keywords.
- Lead statuses, contact fields, plan & usage, change password, sandbox message simulator for testing.

5.12 Super Admin panel

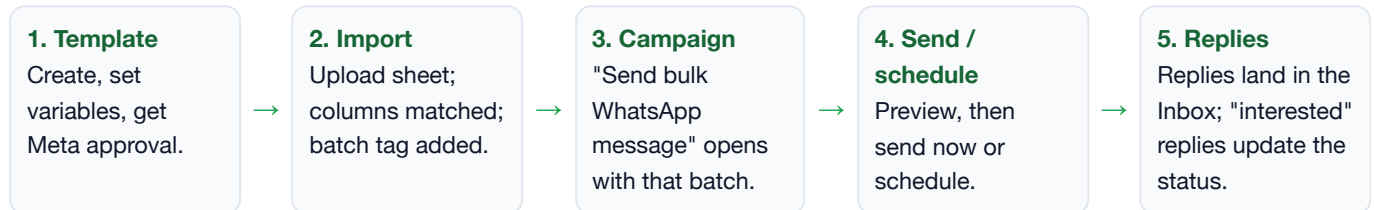
- Dashboard: total and active businesses, suspended businesses, plans, monthly recurring revenue, usage.
- Businesses: create (with admin login), edit, suspend/activate, renew subscription (+N months), change plan, login as admin, reset admin password, delete (with name confirmation).
- Plans: price, limits (agents, contacts, messages per month), features, chatbot module, active/inactive.
- Audit logs: who did what, in which business, from which IP.

6. Everyday workflows

A. Lead from a Facebook ad to a sale



B. Bulk offer from an Excel sheet



C. Onboarding a new client business (Super Admin)

1. Super Admin → Businesses → New business: name, plan, months, admin name/email/password.
2. Admin logs in, opens Settings → WhatsApp number and enters Phone number ID, WABA ID and a permanent token (or Super Admin helps via "Login as admin").
3. Admin adds agents, lead statuses, contact fields, templates and the chatbot.
4. Super Admin records each month's payment with "Renew". If unpaid, sending stops at expiry, or the business can be suspended.

D. A normal day for an agent

1. Open Dashboard → see own open chats and follow-ups due today.
2. Inbox → reply to unread chats; the bot has already answered basic questions.
3. Update each lead's status from the chat header, add tags/notes, set the next follow-up.
4. Resolve finished chats; if the customer writes again, the chat reopens automatically.

7. Business model & how to earn from it

The product is built as **SaaS (software as a service)**: you host it once and charge each business a monthly subscription. Each business connects its own WhatsApp number and pays Meta directly for its WhatsApp message charges, so your income is mostly profit after hosting costs.

Revenue streams

Stream	How it works
Monthly subscription	Main income. Plans with limits on agents, contacts and messages (Starter / Growth / Business, editable by the Super Admin). Renewed monthly from the Super Admin panel.
Setup / onboarding fee	One-time charge for Meta Business verification help, number connection, template writing, chatbot setup and importing old leads.
Annual plans	Offer 10 months' price for 12 months; renew with +12 months. Improves cash flow and retention.
Add-ons	Extra agents, more monthly messages, chatbot module (can be switched per plan), campaign management as a service.
Done-for-you services	Running campaigns, writing templates, managing Click-to-WhatsApp ads for clients, training their staff.
White-label / reseller	Agencies resell the platform to their own clients (a roadmap item for custom branding).

Example earnings (illustration)

Using the default plan prices from the seed data (₹999 Starter, ₹2,499 Growth, ₹5,999 Business per month). Your real prices are set in Super Admin → Plans.

Client mix	Monthly revenue	Yearly revenue
10 Starter + 10 Growth	₹34,980	₹4,19,760
20 Growth + 5 Business	₹79,975	₹9,59,700
50 Growth + 10 Business	₹1,84,940	₹22,19,280

Main running costs

- Server (VPS / cloud) and a managed MongoDB (e.g. MongoDB Atlas). A small setup can serve dozens of businesses; scale up as usage grows.
- Domain and SSL, backups, file storage for media.
- Support time for onboarding and Meta approvals.
- WhatsApp message charges are billed by Meta to each business's own WhatsApp account, not to you, unless you choose to resell messages.

Tips to protect profit: keep plan limits realistic (message limits protect your server and the client's Meta bill); collect payment before "Renew"; use "Suspend" for non-payment; charge a setup fee because Meta verification takes effort; offer annual plans.

How to sell it: start with one niche (e.g. coaching institutes in your city), demo the chatbot and ad-lead tracking in sandbox mode, sell the outcome ("no missed enquiries, know which ad brings admissions"), and offer a 7–14 day trial with help on the first campaign.

8. How businesses profit from using it

Benefit	Why it increases revenue or saves cost
Faster first reply	Chatbot answers instantly, 24×7; agents get alerts. Faster replies convert more enquiries.
No lost leads	Every enquiry is saved with a status and follow-up date; nothing depends on one phone.
Better ad spend	See which Facebook/Instagram ad brings leads and conversions; stop ads that don't work.
Repeat sales	Bulk campaigns to "Interested" or past customers for new batches, offers and festivals.
Less staff time	Bot handles repeated questions; bulk messaging replaces manual forwarding; shared inbox avoids duplicate work.
Accountability	Team report shows chats, reply time and conversions per agent.
No ban risk	Uses Meta's official API and approved templates, unlike unofficial bulk-sender tools.
Owns its data	All leads in one place, exportable to Excel any time.

9. Plans, limits & billing

Plans are created and edited by the Super Admin. Default plans (from the seed data):

Plan	Price / month	Agents*	Contacts	Messages / month	Highlights
Starter	₹999	2	1,000	2,000	Team inbox, bulk campaigns, CSV import
Growth	₹2,499	5	10,000	20,000	+ Auto-assignment, campaign scheduling
Business	₹5,999	20	1,00,000	1,00,000	+ Priority support

* Agents exclude the business admin. The chatbot module can be enabled or disabled per plan.

How billing works in the product

- Each business has a subscription with a status (trial / active / cancelled) and an end date.
- Super Admin records a payment with **Renew (+N months)**, or changes the plan.
- When the subscription ends or the business is suspended, sending stops (chats can still be read), and running campaigns pause.
- Limits are enforced: adding agents or contacts beyond the plan is blocked; outgoing messages are counted each calendar month and campaigns pause at the limit.
- Online payment collection (Razorpay/Stripe) is on the roadmap; today payments are collected outside the app and recorded by the Super Admin.

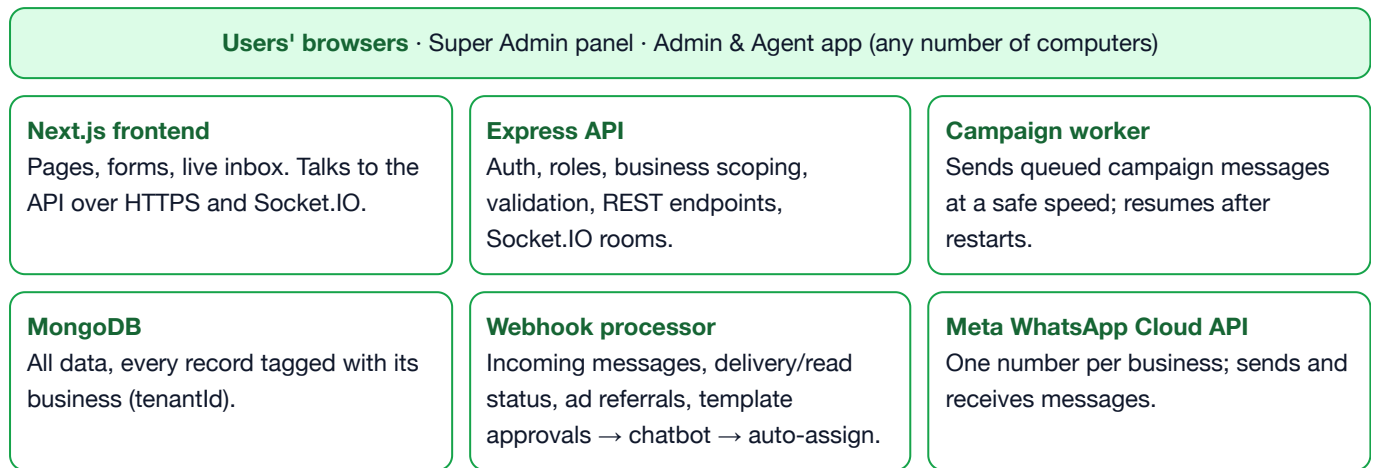
10. Technology stack

Layer	Technology	Used for
Frontend	Next.js 16 (App Router), React 19	All panels (business app and Super Admin) as a fast single-page web app.
Styling & icons	Tailwind CSS 4, Lucide icons	Responsive UI that works on desktop, tablet and phone.
Backend	Node.js 20, Express 5	REST API, WhatsApp webhook, campaign worker, file uploads.
Database	MongoDB with Mongoose 9	Businesses, users, contacts, chats, messages, templates, campaigns, chatbot, audit logs.
Real-time	Socket.IO 4	Live inbox, new-message alerts, campaign progress, template status, on every open computer.
WhatsApp	Meta WhatsApp Cloud API (Graph API) via axios	Sending text, media, templates, interactive buttons/lists; receiving messages, statuses, ad referrals, template approvals by webhook.
Validation	Zod 4	Checks every request body, so bad data is rejected with clear messages.
Auth & security	JWT, bcryptjs, helmet, express-rate-limit, Node crypto (AES-256-GCM)	Logins, password hashing, secure headers, brute-force protection, encrypted WhatsApp tokens.
Files & sheets	multer, ExcelJS, csv-parse	Media uploads, Excel/CSV import and export.
Logging	morgan + audit log collection	Request logs and a permanent record of important actions.
Dev database	mongodb-memory-server (local only)	Runs a local MongoDB automatically in development when no MONGO_URI is set.

Code structure

```
whatsapp-crm/
├── server/                Node.js + Express API
│   ├── src/
│   │   ├── index.js      app setup, routes, production frontend serving
│   │   ├── config/       environment settings
│   │   ├── middleware/   login (JWT), roles, business scoping
│   │   ├── models/       Tenant, User, Plan, Contact, Conversation, Message,
│   │   │                   Template, Campaign, CampaignRecipient, Chatbot, AuditLog
│   │   ├── routes/       auth, superadmin, dashboard, team, contacts, conversations,
│   │   │                   templates, campaigns, chatbot, settings, webhook, misc
│   │   ├── services/     whatsapp, messaging, webhookProcessor, campaigns (worker),
│   │   │                   chatbot, leadStatuses, contactFields, sheets, socket, audit,
│   │   │                   subscription
│   │   └── seed.js        super admin, plans, demo business
├── web/                  Next.js frontend
│   ├── app/login, app/super-admin/..., app/app/... (dashboard, inbox, contacts,
│   │   campaigns, templates, chatbot, team, settings)
│   ├── components/       UI kit, shell, inbox, contacts import wizard, shared parts
│   └── lib/               API client, auth, socket, formatting, lead statuses, fields
```

11. Architecture



Key design points

- **Multi-tenant isolation:** every business-scoped record has a `tenantId` ; all business routes pass through a guard that filters by the logged-in user's business.
- **Real-time rooms:** admins of a business, each agent and each user have their own Socket.IO room, so agents only receive events for chats they may see.
- **Incoming message flow:** Meta webhook → signature check → find business by phone number ID → save contact/message (+ ad details, auto lead status, opt-out) → push to screens → chatbot → round-robin assignment.
- **Campaign flow:** audience snapshotted into recipient rows → worker claims each row atomically → sends template → Meta status webhooks update delivered/read counts.
- **Sandbox mode:** without Meta credentials the system simulates sending, delivery and read receipts so everything can be tested.
- **Production:** one Node process serves both the API (`/api`) and the built Next.js frontend on the same domain.

12. Security & data protection

Area	Protection
Passwords	Hashed with bcrypt; never stored in plain text. Minimum 8 characters.
Login	JWT tokens; login rate limit (20 attempts per 15 minutes per IP) against password guessing.
Roles	Every API route checks the role (Super Admin / Admin / Agent) and the business.
WhatsApp tokens	Stored encrypted with AES-256-GCM using a server key (<code>ENCRYPTION_KEY</code>).
Webhook	Meta's <code>X-Hub-Signature-256</code> is verified with the App Secret; the verify token protects setup.
Input	Every request validated with Zod; upload size limits (10 MB sheets).
HTTP	Secure headers (helmet), CORS restricted to the app's URL.
Support access	"Login as admin" is time-limited (2 h), shown in a banner and written to the audit log.
Deletes	Messages are soft-deleted (kept for audit); business deletion requires typing the business name.
Audit log	Logins-as, settings changes, campaign actions, template changes, deletions, with user, business and IP.
Customer consent	Opt-out keywords stop marketing messages automatically.

13. WhatsApp rules the product follows

- **Official API only:** uses Meta's WhatsApp Cloud API, so numbers are not banned for using the software (they can still be restricted for spam or poor quality, so send only to people who expect your messages).
- **24-hour customer service window:** after a customer's last message you can send any message for 24 hours. Outside it, only an approved template can be sent. The inbox shows whether the window is open.
- **Templates:** the first message to a new customer and all bulk messages must use Meta-approved templates. Categories: Marketing, Utility, Authentication. Variables like `{{1}}` are filled per customer.
- **Template edits:** an approved template can be edited up to once per 24 hours and 10 times in 30 days; name, language and category cannot change. Edited templates go back to review.
- **Charges:** Meta charges the business for template messages according to its current rate card (it varies by category and country). Replies within the 24-hour window are generally free. Check Meta's pricing page for current rates.
- **Messaging limits:** new numbers can message a limited number of unique customers per day; the limit rises as quality stays high and volume grows.
- **Opt-out:** customers who reply STOP are excluded from campaigns.

14. Setup, going live & deployment

Run locally

```
# Backend
cd server
npm install
cp .env.example .env          # set JWT_SECRET and ENCRYPTION_KEY
npm run seed                  # super admin, 3 plans, demo business
npm run dev                   # http://localhost:4000

# Frontend (new terminal)
cd web
npm install
echo "NEXT_PUBLIC_API_URL=http://localhost:4000" > .env.local
npm run dev                   # http://localhost:3000
```

Server environment variables

Variable	Purpose
NODE_ENV , PORT , CLIENT_URL	Mode, API port, the app's public URL (for CORS).
MONGO_URI	MongoDB connection (empty in development = automatic local database).
JWT_SECRET , JWT_EXPIRES_IN	Login token signing and lifetime.
ENCRYPTION_KEY	64 hex characters; encrypts WhatsApp access tokens.
WA_GRAPH_VERSION , WA_APP_SECRET , WA_WEBHOOK_VERIFY_TOKEN	Meta Graph API version, app secret (webhook signature) and webhook verify token.
CAMPAIGN_MPS , CAMPAIGN_WORKER_INTERVAL_MS	Bulk sending speed (messages per second) and worker tick.
SEED_SUPERADMIN_EMAIL , SEED_SUPERADMIN_PASSWORD , SEED_DEMO_PASSWORD	Logins created by <code>npm run seed</code> .

Connecting a real WhatsApp number

1. Meta Business account + Meta developer app with the WhatsApp product; add and verify the business phone number (not active on the WhatsApp app); add a payment method in WhatsApp Manager.
2. Meta app → WhatsApp → Configuration → Webhook: Callback URL `https://<your-domain>/api/webhook/whatsapp` , Verify token = `WA_WEBHOOK_VERIFY_TOKEN` ; subscribe to **messages** and **message_template_status_update**.
3. Put the Meta App Secret in `WA_APP_SECRET` .
4. Create a System User with a permanent token (whatsapp_business_messaging, whatsapp_business_management).
5. In the CRM, Admin → Settings → WhatsApp number: enter Phone number ID, WABA ID and the token → the business switches to live mode.

Production checklist

- A server (VPS or cloud) with Node.js 20+, a domain and HTTPS (needed by Meta's webhook).
- Managed MongoDB (e.g. Atlas) with backups; set `MONGO_URI` .
- Strong unique `JWT_SECRET` and `ENCRYPTION_KEY` ; keep `.env` secret.
- Build the frontend, then start the backend in production mode (it serves both):

```
cd web && npm install && NEXT_PUBLIC_API_URL= npm run build  
cd ../server && npm install && NODE_ENV=production npm start
```

- Run with a process manager (e.g. PM2 or systemd) so it restarts automatically; point the Meta webhook to the domain (a permanent URL, unlike a temporary tunnel).
- Change the seed passwords after the first login.

15. Current limitations & roadmap

Item	Status / note
Online payments (Razorpay / Stripe) and auto-renewal	Planned. Today payments are recorded manually with "Renew".
Facebook Lead Ads forms sync	Planned. Needs extra Meta app permissions and app review. Click-to-WhatsApp ads are fully supported.
Meta Conversions API (send "Converted" back to Meta)	Planned. Needs Meta app review.
Media / button templates	Planned. Text templates with header, body, footer and variables are supported.
AI chatbot from the business's FAQ	Planned. The rule-based chatbot is available now.
Email notifications, password reset by email	Planned. Admins reset agent passwords; Super Admin resets admin passwords.
Object storage (S3) for media	Planned for scale. Media is stored on the server today.
Duplicate campaign, bulk remove tags, white-label branding	Planned.
Editing a running or completed campaign	By design not allowed (messages already sent); use pause/cancel or a new campaign.

16. Glossary

Term	Meaning
Business / Tenant	One company using the platform, with its own number, team and data.
Contact / Lead	A customer or prospect who messaged the business or was imported.
Lead status	Where the lead is in the sales process (New, Interested, Converted...).
Template	A message format pre-approved by Meta, required for first messages and bulk sends.
Variable {{1}}	A blank in a template filled per customer (e.g. their name) when sending.
24-hour window	Time after the customer's last message when any free-form reply is allowed.
Campaign	One bulk send of a template to an audience.
WABA	WhatsApp Business Account at Meta that holds the phone number and templates.
Phone number ID	Meta's ID for the business number, used by the API.
Webhook	The URL where Meta sends incoming messages and status updates to the CRM.
Click-to-WhatsApp ad	A Facebook/Instagram ad whose button opens a WhatsApp chat with the business.
Opt-out	A customer who asked not to receive bulk messages (e.g. by replying STOP).
Sandbox mode	Testing mode that simulates WhatsApp without sending real messages.
Impersonation (login as admin)	Super Admin opening a business as its admin for support, logged in the audit log.
Round-robin	Giving new chats to agents one by one in turn.

WhatsApp CRM · Product documentation v1.0 · October 2026. WhatsApp and Meta rules and prices change over time; check Meta's official documentation for the latest limits and rates.